

Theorem (Triangle Inequality). *Let $(V, \langle \cdot, \cdot \rangle)$ be an inner-product space and let $x, y \in V$. Then*

$$\|x + y\| \leq \|x\| + \|y\|.$$

Proof. **1. Norm expressed through the inner product.** By definition of the norm induced by the inner product,

$$\|v\|^2 = \langle v, v \rangle \quad \text{for every } v \in V. \quad (1)$$

2. Expand $\|x+y\|^2$. Using bilinearity in the first argument and conjugate-linearity in the second argument we obtain

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle. \end{aligned}$$

Since $\langle y, x \rangle = \overline{\langle x, y \rangle}$, the sum of the mixed terms equals twice the real part of $\langle x, y \rangle$:

$$\langle x, y \rangle + \langle y, x \rangle = \langle x, y \rangle + \overline{\langle x, y \rangle} = 2 \operatorname{Re} \langle x, y \rangle. \quad (3)$$

Substituting (3) into (2) and using (1) for $\|x\|^2$ and $\|y\|^2$ gives

$$\|x + y\|^2 = \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2. \quad (4)$$

3. Bound the real part of the inner product. The Cauchy–Schwarz inequality states

$$|\langle x, y \rangle| \leq \|x\| \|y\|. \quad (5)$$

Because the real part of any complex number does not exceed its modulus,

$$\operatorname{Re} \langle x, y \rangle \leq |\langle x, y \rangle|. \quad (6)$$

Combining (5) and (6) we obtain

$$\operatorname{Re} \langle x, y \rangle \leq \|x\| \|y\|. \quad (7)$$

4. Insert the bound into (4). From (4) and (7),

$$\|x + y\|^2 \leq \|x\|^2 + 2 \|x\| \|y\| + \|y\|^2 = (\|x\| + \|y\|)^2. \quad (8)$$

5. Take square roots. Norms are non-negative, i.e. $\|z\| \geq 0$ for all $z \in V$. Hence both sides of (8) are non-negative, and the monotonicity of the square-root function on $[0, \infty)$ yields

$$\|x + y\| \leq \|x\| + \|y\|. \quad (9)$$

Equality case. Equality in (??) occurs precisely when equality holds in both (??) and (??). Equality in the Cauchy–Schwarz inequality holds iff x and y are linearly dependent, i.e. $y = \lambda x$ for some $\lambda \in \mathbb{R}$. Equality in (??) holds iff $\langle x, y \rangle$ is a non-negative real number, which forces $\lambda \geq 0$.

Consequently, equality in the triangle inequality holds exactly when there exists a non-negative real scalar λ such that $y = \lambda x$ (including the trivial cases $x = 0$ or $y = 0$). $\square$